Firm Wage Setting, On-the-Job Search, and the Inflationary Effects of Supply Shocks: An AD-AS Framework with Quits and Wage Inflation (Not for Publication)

Justin Bloesch[*] Seung Joo Lee[†] Jacob P. Weber[‡]

August 6, 2026

Abstract

We present a simplified version of Bloesch, Lee and Weber (2026) using aggregate demand (AD) and aggregate supply (AS) curves.

1 Simplified Model

This section describes a two-period special case of the wage-posting model in Bloesch, Lee and Weber (2026) where the economy responds to shocks in period $t = 0$ under the assumption that it returns to steady state after $t > 0$. This simplified model allows us to demonstrate that the economy can be described using aggregate demand (AD) and aggregate supply (AS) curves.

For simplicity and tractability, we assume no exogenous separations ($s = 0$); linear vacancy posting costs ($\chi = 0$); that unemployed and employed workers receive equal consumption from the household ($\xi = 1$); and that all employed workers search each period (i.e., $\lambda_{EE} = 1$) so that

[*]Department of Economics and ILR, Cornell University. Email: jb2722@cornell.edu
[†]Saïd Business School, University of Oxford. Email: seung.lee@sbs.ox.ac.uk
[‡]Federal Reserve Bank of New York. Email: jake.weber@ny.frb.org

The views expressed in this paper are those of the authors and do not necessarily reflect the position of the Federal Reserve Bank of New York or the Federal Reserve System.

JEL Codes: E31, E52.
Keywords: Monopsony, Inflation, Cost-of-living shocks, On-the-job search.

tightness θ_t becomes equal to V_t, the number of vacancies posted. Also, we now assume that there is one sector so that $C_t = Y_t$,[1] so that $\Pi_{Y,t}$ now describes aggregate price inflation in the economy. Finally, we assume that the price setting of firms is fully flexible (i.e., $\psi = 0$) while wages remain sticky, so that there continues to be a role for monetary policy in the model.

Up to a first order, the following equations (derived in Appendix A) describe the model at $t = 0$ in log-deviation from the steady state:

Monetary Policy Rule (i.e., Taylor rule) $i_0 = \rho + \phi\check{\Pi}_{Y,0} + \check{\Pi}_{Y,1} + \check{\varepsilon}_0$

Euler Equation: $\check{C}_0 = \rho - i_0 + \check{\Pi}_{Y,1}$

Production Function: $\check{C}_0 = \check{A}_0 + \check{N}_0$

Law of Motion for Employment: $\check{N}_0 = \omega_V \check{V}_0$

Firm Price + Wage Optimality Conditions: $\check{\Pi}_{Y,0} = -\Omega_A \check{A}_0 + (\Omega_V + \phi_V)\, \check{V}_0$

where N is employment, V is vacancies posted, Π_Y is price inflation, and output Y, consumption C, and employment N are related $C_t = Y_t = A_t N_t$, so that $\check{A}_0$ is a productivity shock and $\check{\varepsilon}_0$ is a contractionary monetary policy shock. To interpret these equations, first note that all coefficients $\Omega_A, \Omega_V, \phi_V, \omega_V$ are strictly positive.[2]

The first, second and third equations above constitute our aggregate demand relationship. The first equation is a usual Taylor rule, with an adjustment that the monetary authority raises i_0 when the next period service inflation $\check{\Pi}_{Y,1}$ increases.[3] The second states that consumption $\check{C}_0$ increases in discount rate ρ, decreases in policy rate i_0, and increases in next period inflation $\check{\Pi}_{Y,1}$, which are all standard. Combining these equations, we obtain an "Aggregate Demand" (AD) relationship

AD (Monetary Policy Rule + Euler Equation): $\check{N}_0 = -\phi\check{\Pi}_{Y,0} - \check{\varepsilon}_0 - \check{A}_0,$

which states that employment N_0 declines when service inflation $\Pi_{Y,0}$ is high, because the monetary authority raises the real interest rate with a sensitivity $\phi > 0$, and households reduce aggregate consumption demand.

[1] This scenario corresponds to $\eta \to \infty$ and $X_t = 0$ in Bloesch et al. (2026).

[2] In Appendix A.4, we prove that in a steady state with unemployment less than 50%, $\phi_V > 0$.

[3] As the economy returns to the steady state from period $t \geqslant 1$, at period 0, it is an economy with perfect foresight.

The fourth equation is about the relationship between vacancies V_0 and employment N_0: when vacancies V_0 are high, employment N_0 are also high: this is because in our simple model with a unit measure of workers where all unemployed (U) and employed (N) workers search, i.e., $\lambda_{EE} = 1$, we have tightness $\theta_t = \frac{V_t}{U_{t-1}+\lambda_{EE}N_{t-1}} = V_t$. A standard matching function states that a tighter labor market makes it easier for unemployed workers to find jobs, lowering unemployment and raising employment.

Finally, the last equation states that price inflation rises when marginal costs rise: a tight labor market (with higher $\check{V}_0$) means higher turnover costs (thus higher wages), leading to a higher price inflation. With high $\check{A}_0$, workers become more productive, and marginal costs fall, so price inflation drops and the real wage at $t = 0$ rises. Also important is Ω_A, which is a constant related to firms' market power: when firms have a lot of market power, Ω_A is lower, and there is less pass through from a productivity shock that affects marginal costs to inflation $\check{\Pi}_{Y,0}$. High market power implies that firms do not pass on productivity improvements much to customers in the form of lower prices.

AD-AS Representation Using the law of motion for employment to substitute out for $\check{V}_0$ yields the following Aggregate Demand (AD) and Aggregate Supply (AS) curves:

$$\textbf{AD (Monetary Policy Rule + Euler Equation):}\ \check{N}_0 = -\phi\check{\Pi}_{Y,0} - \check{\varepsilon}_0 - \check{A}_0$$

$$\textbf{AS (Firm Prices + Wages + Law of Motion):}\ \check{\Pi}_{Y,0} = -\Omega_A\check{A}_0 + \left(\frac{\Omega_V + \phi_V}{\omega_V}\right)\check{N}_0$$

which allows us to analyze how shocks affect price inflation and employment (equivalently, unemployment).[4]

This AD–AS representation is useful because, to a first-order approximation, changes in employment N serves as a sufficient statistic for changes in labor market tightness V, wage inflation Π^w, and quits rate S. These always move together in the model: quits rate S increase with tightness (θ, or V here when $\lambda_{EE} = 1$) in the model because job-to-job transitions increase when it is more likely that searching workers find a match. Wage inflation increases with tightness (and aggregate employment) because optimizing firms increase in size both by posting vacancies and by offering higher wages to improve the recruiting rate on those vacancies.

[4]The model has no explicit distinction between the unemployed and workers who are out of the labor force. Since we assume that all nonemployed workers search each period, we refer to the nonemployed as unemployed.

2 Impacts of Shocks

Consider the following exercises using Figure 1, which plots these AD-AS curves:

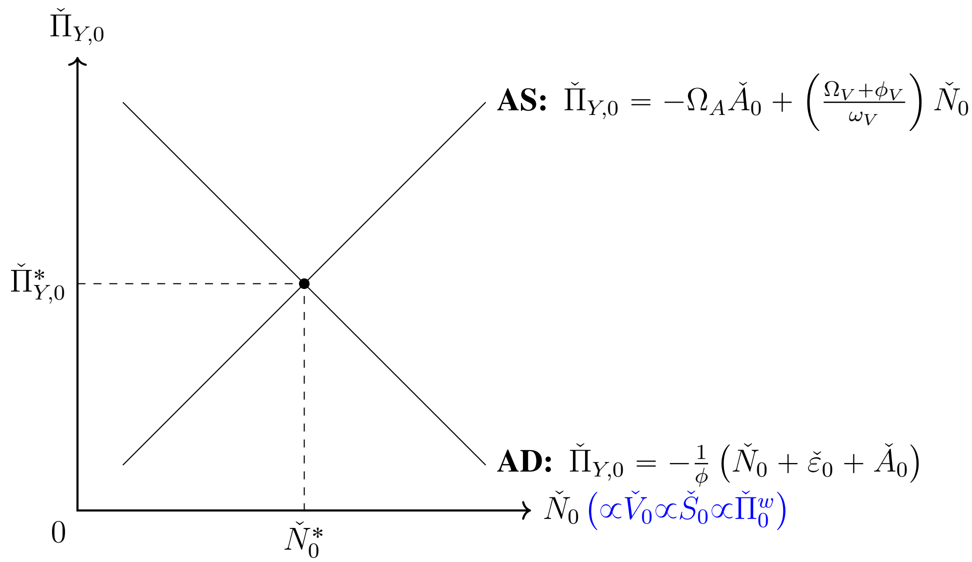
- Contractionary (i.e., positive) monetary policy shock $\check{\varepsilon}_0$ always lowers inflation and employment (and thus vacancy, quits rate, and wage inflation), as seen in Figure 2.
- Total factor productivity (TFP) shocks generate more interesting results. Positive TFP shocks $\check{A}_0 > 0$ always lower prices, but it is not clear what happens to the labor market (employment, tightness, quits, and wage growth) because both AD and AS curves move in response to the shocks:
  - **Very High Taylor Coefficient** ϕ: If monetary policy stabilizes service inflation, $\check{\Pi}_{Y,0}$, sufficiently, then AD curve becomes very flat, i.e., service inflation $\check{\Pi}_{Y,0}$ does not vary much when employment changes. In this case, employment, tightness, quits rate, wage growth will rise (and prices will not move much). This can be understood as follows: when workers become more productive but prices are approximately fixed due to strong monetary policy responses, labor demand will increase, leading to higher real wage, vacancy, employment, tightness, and quits rate. It can be seen in Figure 3. As illustrated in Figure 4, when monetary responsiveness approaches infinity ($\phi \to \infty$), the AD curve becomes completely flat. This implies no change in service inflation $\check{\Pi}_{Y,0}$ and a large increase in employment N_0.
  - **Low Taylor Coefficient** ϕ **and High Market Power (Small** Ω_A**)**: When Ω_A is small in magnitude because, for example, firms' market power is very high, which leads to less pass through from productivity improvements (which come with lower marginal costs) to prices,[5] then AS curve will not shift much. However, with low monetary responsiveness ϕ, positive TFP shocks lower marginal costs, causing deflationary pressures by lowering aggregate demand: when AD falls, service prices $\check{\Pi}_{Y,0}$ fall but wages also fall, leading to decreases in employment, vacancies, quits, and wage inflation, as illustrated in Figure 5.

[5] For example, $\epsilon \simeq 1$ will be such a case, as shown in Appendix A.

– **Central Bank Stabilizing the Labor Market**: Figure 6 shows that when the central bank stabilizes N_0, a positive productivity shock results in a substantial decline in service inflation.

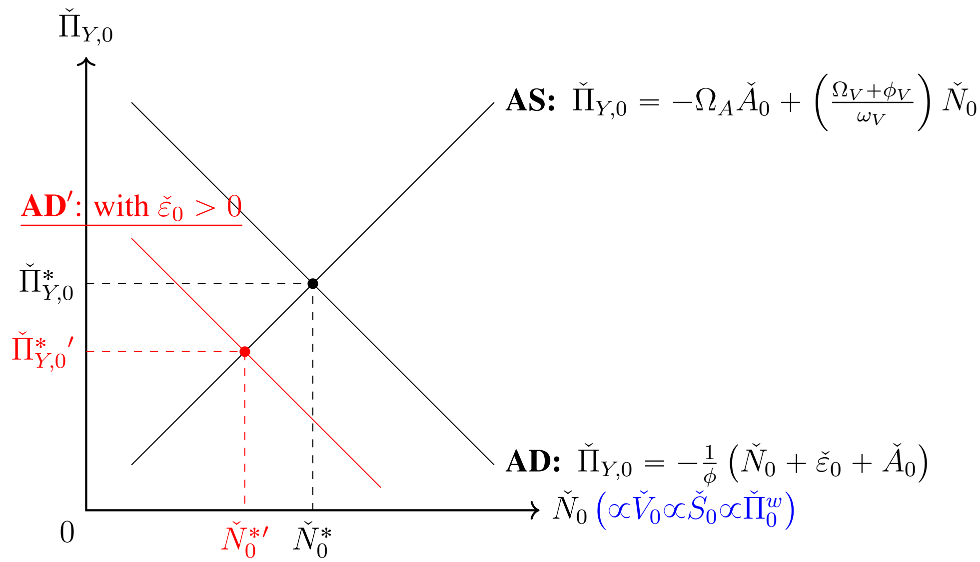
One lesson from our wage-posting model is that, at least to first order, demand (e.g., monetary policy) shocks and TFP shocks affect nominal wage growth only through their effects on labor market tightness. Here, this is summarized by vacancies $\check{V}_0$, but Bloesch, Lee and Weber (2026) show that this holds in the richer dynamic model with a more realistic calibration (wage inflation depends on both deviations in V_t (or quits) and, to a much lesser extent, U_{t-1} from steady state).

Figure 1: Aggregate Demand (AD)-Aggregate Supply (AS) Framework with Quits, Vacancies, and Wage Inflation



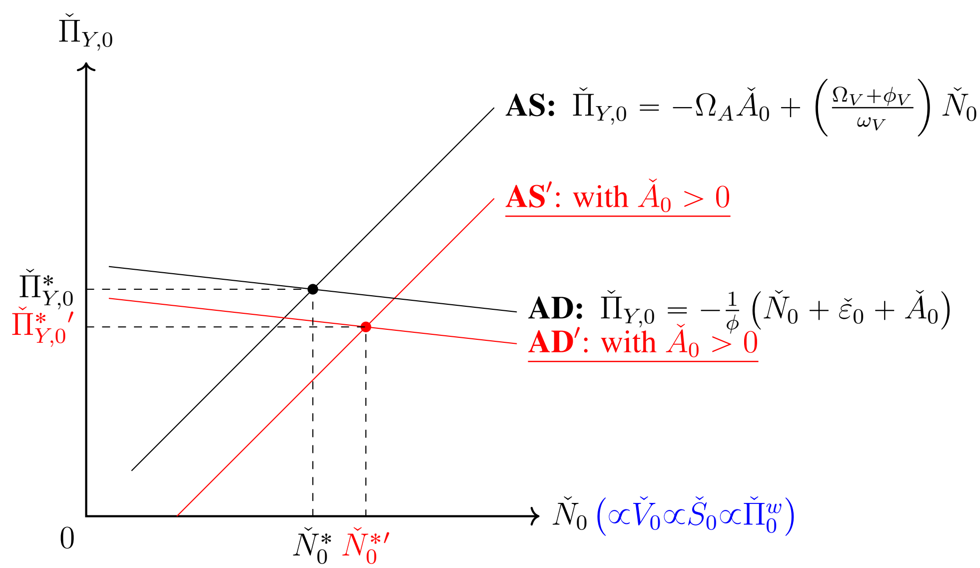
Notes: See text for derivations of the AD and AS curves based on sticky wages and flexible prices. All coefficients $\Omega_A, \Omega_V, \phi_V, \omega_v$ are strictly positive, and $\phi > 0$ governs the strength of the monetary authority's response to inflation.

Figure 2: Monetary Policy Shock $\check{\varepsilon}_0 > 0$



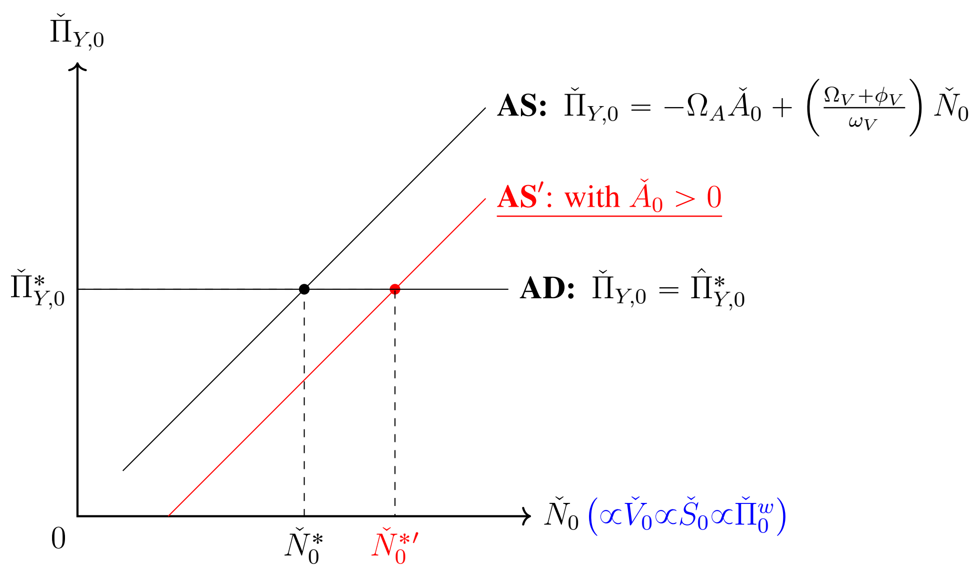
Notes: A contractionary monetary policy shock lowers price inflation (Π_Y) and employment (N), which lowers output. Because changes in N are sufficient statistics for changes in tightness (V), quits (S) and wage inflation (Π^w), these all fall as well.

Figure 3: TFP Shock $\check{A}_0 > 0$ with High Taylor Coefficient $\phi >> 1$



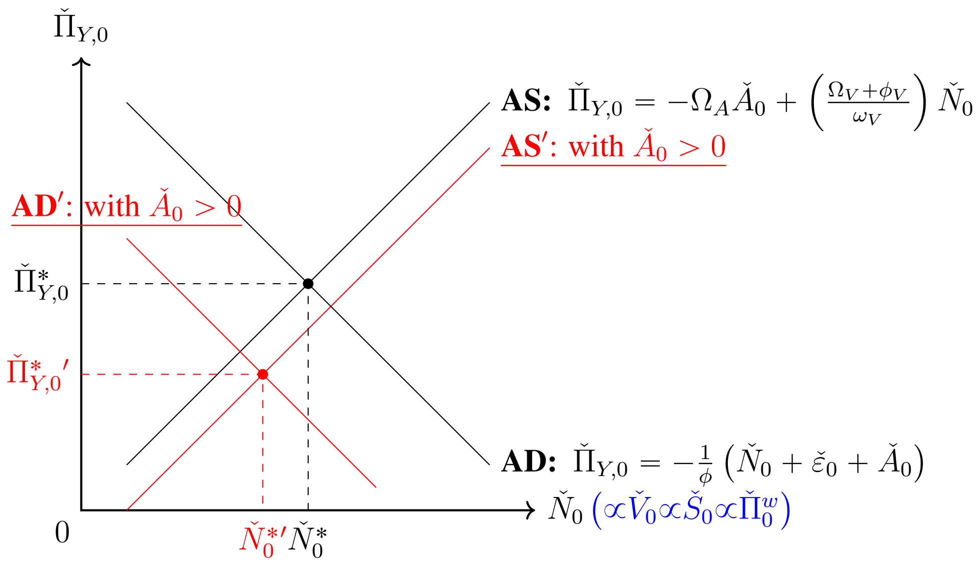
Notes: A positive productivity shock (A_0) raises employment (N) and lowers price inflation (Π_Y), assuming that interest rates respond strongly enough to inflation, so that the AD curve is flat and does not shift much.

Figure 4: TFP Shock $\check{A}_0 > 0$ with Very High Taylor Coefficient $\phi \to \infty$



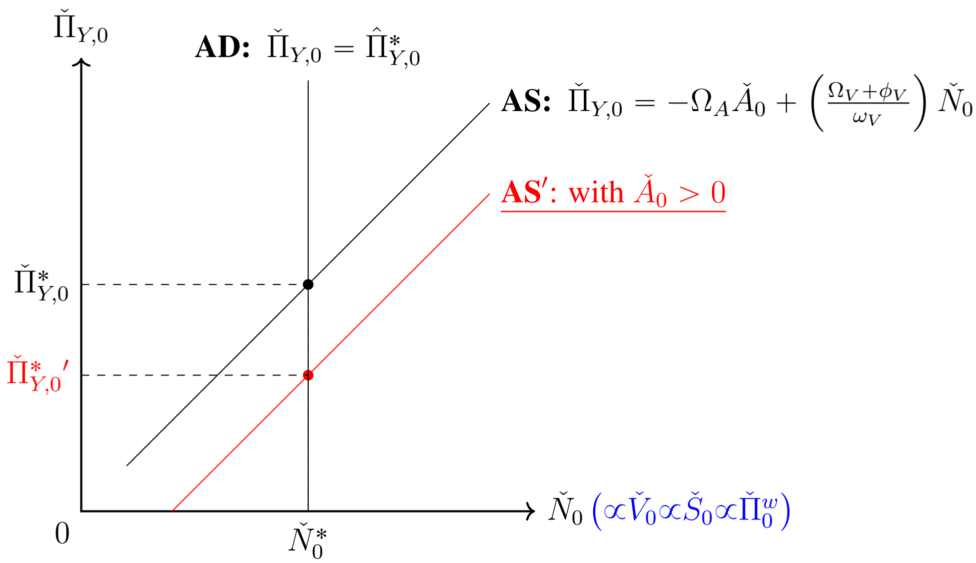
Notes: A positive productivity shock (A_0) raises employment (N) and lowers price inflation (Π_Y), assuming that interest rates respond strongly enough to inflation, so that the AD curve becomes completely flat and does not shift at all.

Figure 5: TFP Shock $\check{A}_0 > 0$ with Low Taylor Coefficient ϕ, High Market Power (Small Ω_A)



Notes: A positive productivity shock can reduce employment (N) and price inflation (Π_Y) if the monetary authority responds weakly to inflation (ϕ is small) and if the AS curve does not shift much because Ω_A is small, which will be the case if firms have a lot of market power and do not pass on productivity improvements to consumers in the form of lower prices. In this case, tightness (V) and quits (S) and wage inflation (Π^W) all fall as well.

Figure 6: TFP Shock $\check{A}_0 > 0$ with Monetary Policy Stbilizing the Labor Market



Notes: A positive productivity shock (A_0) lowers price inflation (Π_Y), assuming that interest rates respond strongly enough to labor market variables N_0, so that the AD curve becomes completely vertical and does not shift at all.

References

Bloesch, Justin, Seung Joo Lee, and Jacob Weber, "Firm Wage Setting, On-the-Job Search, and the Inflationary Effects of Supply Shocks," *Available at SSRN 4734451*, 2026.

Rotemberg, Julio J., "Sticky Prices in the United States," *Journal of Political Economy*, 1982, *90* (6), 1187–1211.

A Derivation for Section 1

Here, we describe how to derive the AD and AS curves described in Section 1 and plotted in Figure 1. We do so by analyzing a two-period special case of the dynamic model developed by Bloesch, Lee and Weber (2026).

In specific, we solve for the economy's response to some shock at $t = 0$ under the assumption that we return to steady state at $t > 0$. As we explained in Section 1, we also assume no exogenous separations (i.e., $s = 0$); that unemployed and employed workers receive the same consumption (i.e., $\xi = 1$); and that all employed workers search (i.e., $\lambda_{EE} = 1$) each period so that tightness θ_t equals V_t, the number of vacancies posted; and fully flexible prices (i.e., $\psi = 0$).

A.1 Firm's Wage-Posting Problem

Perfectly-competitive retailers bundle service types j according to a standard Dixit-Stiglitz production function with an associated ideal price index:

$$\begin{aligned} Y_t &= \left(\int \left(Y_t^j\right)^{\frac{\epsilon-1}{\epsilon}} dj \right)^{\frac{\epsilon}{\epsilon-1}}, \\ P_{y,t} &= \left(\int \left(P_{y,t}^j\right)^{1-\epsilon} dj \right)^{\frac{1}{1-\epsilon}}, \end{aligned}$$

yielding product demand for variety j:

$$\frac{Y_t^j}{Y_t} = \left(\frac{P_{y,t}^j}{P_{y,t}} \right)^{-\epsilon}. \tag{1}$$

The firm j produces only with labor according to production function $Y_t^j = A_t^j N_{jt}$. Firm j sets its nominal wages W_{jt} each period, which is assumed to be the same for all workers in the firm, including new hires. Workers separate from firm j with probability $S(W_{jt}|\{W_{kt}\}_{k\neq j})$ each period, with $S'(W_{jt}|\{W_{kt}\}_{k\neq j}) < 0$: firms retain a higher share of workers each period by paying a higher wage, given other firms' wages. Firms recruit workers by posting vacancies V_{jt}, and the probability that a vacancy successfully results in a hire is $R(W_{jt}|\{W_{kt}\}_{k\neq j})$, with $R'(W_{jt}|\{W_{kt}\}_{k\neq j}) > 0$.[6] The firm pays a linear, per-vacancy hiring cost cW_t to post V_t vacancies, where W_t is the aggregate wage and $c > 0$. Finally, the firm is also subject to wage adjustment frictions à la Rotemberg (1982).

Given this, each firm j maximizes the present discounted value of profits, solving

$$\max_{\substack{\{P_{y,t}^j\},\{Y_t^j\},\\ \{N_{jt}\},\{W_{jt}\},\{V_t^j\}}} \sum_{t=0}^{\infty} \left(\frac{1}{1+\rho}\right)^t \left(P_{y,t}^j Y_t^j - W_{jt}N_{jt} - cV_{jt}W_t - \frac{\psi^w}{2}\left(\frac{W_{jt}}{W_{j,t-1}} - 1\right)^2 W_{jt}N_{jt}\right) \tag{2}$$

subject to the law of motion for employment

$$N_{jt} = (1 - S(W_{jt}))N_{j,t-1} + V_{jt}R(W_{jt}) \tag{3}$$

and the product demand equation (1). From inspecting equations (2) and (3), we can observe that firms in the service sector choose the wage (and other choice variables) taking as given the choices of other firms (embodied in the price index and aggregate output of the service sector), parameters, and the separation and recruiting rates $S(\cdot)$ and $R(\cdot)$ which are decreasing and increasing functions of W_{jt}, respectively.

Equilibrium We characterize a symmetric equilibrium where $P_{y,t}^j = P_{y,t}$, $V_{j,t} = V_t$, $W_{jt} = W_t$, $A_t^j = A_t \; \forall j$. Then defining λ_t as the Lagrange multiplier on the law of motion for employment, the firm's problem yields the following first order conditions:

First-order condition on wages The marginal cost of raising the wage $W_{j,t}$ is equal to $N_{j,t}$, because the wage bill is given by $W_{j,t}N_{j,t}$, plus adjustment costs terms multiplied by ψ^w, which an op-

[6]The dependence of the retention and separation functions, $R(W_{jt}|\{W_{kt}\}_{k\neq j})$ and $S(W_{jt}|\{W_{kt}\}_{k\neq j})$, on wages set by other firms arises from the optimization decisions of households and workers. These optimization problems are detailed in Bloesch, Lee and Weber (2026). For simplicity, we express $R(\cdot)$ and $S(\cdot)$ solely as functions of W_{jt} set by firm j solely.

timizing firm equates with the marginal benefit: the marginal number of new workers $V_t R'(W_t) - N_{t-1}S'(W_t)$ times their shadow value, λ_t as follows:

$$
\begin{aligned}
& N_t + \psi^w(\Pi_t^w - 1)\Pi_t^w N_t + \underbrace{\cancel{\frac{\psi^w}{2}(\Pi_t^w - 1)^2 N_t}}_{\simeq 0} - \frac{1}{1+\rho}\psi^w \left(\Pi_{t+1}^w - 1\right)\left(\Pi_{t+1}^w\right)^2 N_{t+1} \\
& \quad = \lambda_t \left(V_t R'(W_t) - N_{t-1}S'(W_t)\right)
\end{aligned}
$$

where we define the aggregate wage inflation $\Pi_t^w = \frac{W_t}{W_{t-1}}$ and approximate with $(\Pi_t^w - 1)^2 \simeq 0$.

First-order condition on vacancies The marginal (shadow) value of a worker, λ_t, is determined by the first order condition for vacancies: an optimizing firm j chooses vacancies, $V_{j,t}$, so that the marginal value of a new worker is equal to the marginal cost,

$$
\lambda_t = \frac{c}{R(W_t)} W_t .
$$

where the right hand side is the effective marginal cost for firms per unit hire.

First-order condition on prices For simplicity, we assume that service prices are fully flexible.[7] An optimizing firm sets the price so that the marginal revenue of a worker equals a standard markup over marginal cost of that worker, which includes wage-adjustment costs (though these effects are second-order) and also the recruitment costs

$$
P_{y,t}A_t = \frac{\epsilon}{\epsilon - 1}\left(W_t + \cancel{\frac{\psi^w}{2}\underbrace{(\Pi_t^w - 1)^2}_{\simeq 0} W_t A_t N_t} + \lambda_t - \frac{1}{1+\rho}\lambda_{t+1}(1 - S(W_{t+1})) \right).
$$

A.2 Closing the Model: Households, Workers, and Monetary Policy

There are a unit mass of workers so that $N_t + U_t = 1$. The household and workers' problem is fully described in Bloesch, Lee and Weber (2026). In brief, households tax and provide unemployment benefits to workers (either employed or unemployed) while trading zero-net supplied bonds such that aggregate consumption follows a standard Euler equation. Workers choose between job offers

[7] In Bloesch, Lee and Weber (2026), we assume that prices are sticky à la Rotemberg (1982) as well.

when they arrive: both unemployed and employed workers search each period and match with a firm with probability $f(\theta_t)$, which is an increasing function of market tightness θ_t, or the number of job openings divided by the number of searchers, which simplifies here when all employed and unemployed workers search to just $\theta_t = \frac{V_t}{\lambda_{EE} N_{t-1} + U_{t-1}} = V_t$.[8]

Since workers have a time-varying, idiosyncratic utility associated with working at a particular firm, we have job-to-job quits and endogenous quits into unemployment even though the household transfer scheme ensures that workers never have higher consumption in unemployment: the household's tax and transfer scheme fixes consumption to be equal in both the employed and unemployed states; we also rule out exogenous separations here, so that we describe separations S_t as quits, which include job-to-job quits and quits into unemployment.[9]

With these assumptions, the law of motion for employment characterizes "labor supply" as an increasing function of tightness, which is here equivalent to vacancies posted:

$$N_t = (1 - S(V_t))N_{t-1} + V_t R(V_t)$$

where the separation and recruiting rates, $S(\cdot)$ and $R(\cdot)$, which arise from the workers' problem, are defined in Bloesch, Lee and Weber (2026), who show that in a symmetric equilibrium these two no longer depend on the wage, but are instead solely functions of labor market tightness (here, V_t): with our simplifying assumptions, these functions become

$$R(V_t) = \frac{g(V_t)}{2}, \tag{4}$$

$$S(V_t) = \frac{f(V_t) + \lambda_{EU}}{2}, \tag{5}$$

where $g(V_t)$ is a decreasing function. To see that N_t is increasing in V_t, note the our definitions for

[8]Note that this is a special case of the model in Bloesch, Lee and Weber (2026) where all employed workers are allowed to search each period ($\lambda_{EE} = 1$) so that labor market tightness $\theta_t = \frac{V_t}{\lambda_{EE} N_{t-1} + U_{t-1}} = V_t$.

[9]Formally, this exposition sets exogenous quit rate $s = 0$ and the fixed consumption ratio for the employed over the unemployed targeted by the household to be $\xi = 1$ in Bloesch, Lee and Weber (2026), who consider the general case where $\xi, s \geqslant 0$ as well as alternative modeling assumptions where the household does not fix the consumption of employed vs. unemployed households.

$g(V_t) = \frac{1}{(1+V_t^2)^{\frac{1}{2}}}$ and $f(V_t) = V_t g(V_t)$ imply that

$$N_t = S(V_t)\,(1 - N_{t-1}) + N_{t-1} - \frac{\lambda_{EU}}{2} \tag{6}$$

which is increasing in tightness V_t because both $f(V_t)$ and quits $S(V_t)$ are increasing in tightness V_t. This equation makes a strong, but intuitive, prediction: when the labor market is hot, vacancies, quits and employment will all be jointly high.

Next, we characterize *labor demand* from the firm's side. Based on the first-order condition for vacancies to substitute out for λ_t, we can find that the real wage $\bar{W}_t = \frac{W_t}{P_{y,t}}$ is determined by the first-order condition for prices, which simplifies to:

$$\bar{W}_t = \left(1 + \underbrace{\frac{c}{R(V_t)}}_{=\frac{\bar{\lambda}_t}{\bar{W}_t}}\right)^{-1} \left[\frac{\epsilon - 1}{\epsilon} A_t + \frac{(1 - S(V_{t+1}))\Pi_{Y,t+1}}{1+\rho}\bar{\lambda}_{t+1}\right]$$

where $\bar{\lambda}_{t+1} \equiv \frac{\lambda_{t+1}}{P_{y,t+1}}$. When the labor market is tight, λ_t is high, implying marginal costs are high, which translates into a lower *real* wage $\bar{W}_t$. In log differences, this equation links wage inflation and price inflation to changes in marginal costs.

We close the model by assuming the monetary authority reduces aggregate demand whenever current and/or expected inflation is high. By combining the Euler equation with log utility and the nominal interest rate rule $1 + i_t = (1+\rho)\Pi_{Y,t}^{\phi}\Pi_{Y,t+1}\varepsilon_t$, we obtain

$$\frac{1}{C_t} = \frac{\Pi_{Y,t}^{\phi}\varepsilon_t}{C_{t+1}}$$

which shows that consumption $C_t = Y_t = A_t N_t$ is decreasing in current inflation.[10] Note that an increase in ε_t is a contractionary monetary policy shock.

To see what happens to wage inflation, we use the first-order condition for wages—the wage Phillips Curve—which writes wage inflation as a function of labor market tightness: again substi-

[10]This is a one-sector version of Bloesch, Lee and Weber (2026) which sets $C_t = Y_t$, with no endowment good.

tuting for λ_t, and dropping terms approximately zero, we can write

$$1 + \psi^w(\Pi^w_t - 1)\Pi^w_t = \frac{\lambda_t}{N_t}\left(V_t R'(W_t) - N_{t-1}S'(W_t)\right) + \frac{1}{1+\rho}\psi^w\left(\Pi^w_{t+1} - 1\right)\left(\Pi^w_{t+1}\right)^2\frac{N_{t+1}}{N_t}$$

The right hand side of the above equation is generally increasing in tightness V_t: up to a first order, Bloesch, Lee and Weber (2026) show that generically in this model, we can write

$$\check{\Pi}^w_t = \phi_V \check{V}_t + \phi_U \check{U}_{t-1} + \frac{1}{1+\rho}\check{\Pi}^w_{t+1} \tag{7}$$

where the "check" variables denote log deviation from steady state, and where $\phi_V > 0$.[11]

A.3 Monetary Policy and TFP Shocks at $t = 0$

To a first-order approximation, the following equations characterize the model. Technically, we should substitute $\bar{\lambda}_{t+1}$ using the first-order condition for vacancies, but for the moment we keep it in the system.

Wage Phillips Curve: $\check{\Pi}^w_t = \phi_V \check{V}_t + \phi_U \check{U}_{t-1} + \frac{1}{1+\rho}\check{\Pi}^w_{t+1}$

Law of Motion for Employment: $\check{N}_t = \frac{S(1-N)}{N}\check{S}_t + (1-S)\check{N}_{t-1}$

Monetary Policy Rule + Euler Equation: $\check{N}_t + \check{A}_t = -\phi\check{\Pi}_{Y,t} - \check{\varepsilon}_t$

Pricing Equation: $\check{\bar{W}}_t = \Omega_A \check{A}_t - \Omega_V \check{V}_t - \Omega_S \check{S}_{t+1} + \Omega_\lambda \check{\bar{\lambda}}_{t+1} + \Omega_\Pi \check{\Pi}_{Y,t+1}$

where the last equation comes from the log-linear form of

$$\bar{W}_t = \left(1 + \frac{c}{R(V_t)}\right)^{-1}\left[\frac{\epsilon - 1}{\epsilon}A_t + \frac{(1 - S(V_{t+1}))\Pi_{Y,t+1}}{1+\rho}\bar{\lambda}_{t+1}\right]$$

[11] When $\chi = 0$ (i.e., linear vacancy costs as we assume in this section) rather than $\chi = 1$ in Bloesch et al. (2026), it becomes difficult to demonstrate this result in full generality. However, the result holds for any realistic steady state. In particular, it can be shown that a sufficient—though not necessary—condition for $\phi_V > 0$ is that the steady-state unemployment rate is below 50%. See Appendix A.4 for details.

and note from equation (5) we also have

$$\check{S}_t = \underbrace{\frac{f'(V)V}{f(V)+\lambda_{EU}}}_{>0}\check{V}_t.$$

Now, assume that the economy is in steady state when an unanticipated shock hits at $t = 0$. If we assume a return to steady state in $t = 1$ then letting

$$\omega_V \equiv \frac{S(1-N)}{N}\left(\frac{f'(V)V}{f(V)+\lambda_{EU}}\right) > 0,$$

the following must hold:

Wage Phillips Curve: $\check{\Pi}_0^w = \phi_V \check{V}_0$

Law of Motion for Employment: $\check{N}_0 = \omega_V \check{V}_0$

Monetary Policy Rule + Euler Equation: $\check{N}_0 = -\phi\check{\Pi}_{Y,0} - \check{\varepsilon}_0 - \check{A}_0$

Pricing Equation: $\check{\check{W}}_0 = \Omega_A \check{A}_0 - \Omega_V \check{V}_0$

Combining the wage Phillips curve and pricing equation yields the following: note the first equation has $\check{\Pi}_0^w = \ln W_0 - \ln W = \check{W}_0 = \phi_V \check{V}_0$, so that we can write:

$$\check{\Pi}_0^W - \check{\Pi}_{Y,0} = \Omega_A \check{A}_0 - \Omega_V \check{V}_0$$

and

Law of Motion for Employment: $\check{N}_0 = \omega_V \check{V}_0$

Monetary Policy Rule + Euler Equation: $\check{N}_0 = -\phi\check{\Pi}_{Y,0} - \check{\varepsilon}_0 - \check{A}_0$

Pricing + Wage Phillips Curve: $\check{\Pi}_{Y,0} = -\Omega_A \check{A}_0 + (\Omega_V + \phi_V)\check{V}_0$

Note that this last equation is intuitive: prices rise when marginal costs rise: a tight labor market means higher turnover costs (and higher wages) which is why this is increasing in $\check{V}_0$ and decreasing in $\check{A}_0$. Then using the law of motion for employment to substitute out of for $\check{V}_0$, we derive the

AD and AS curves in Figure 1:

$$\textbf{AD (Monetary Policy Rule + Euler Equation):}\ \check{N}_0 = -\phi\check{\Pi}_{Y,0} - \check{\varepsilon}_0 - \check{A}_0$$

$$\textbf{AS (Firm Prices + Wages + Law of Motion):}\ \check{\Pi}_{Y,0} = -\Omega_A\check{A}_0 + \left(\frac{\Omega_V + \phi_V}{\omega_V}\right)\check{N}_0$$

A.4 Sufficient Condition for $\phi_V > 0$ in Equation (7)

In the section, we claimed that in a steady state with unemployment $U < 50\%$, $\phi_V > 0$. We justify this claim here. Consider the following definition for ϕ_V from the Appendix of Bloesch, Lee and Weber (2026): the coefficient on vacancies in the wage Phillips curve is defined as:

$$\phi_V \equiv \frac{\kappa}{\psi_w}\left(\Lambda_1 + \Delta_1(g_{S,V} - g_{R,V}) - \epsilon_S g_{\epsilon_S,V}\right)$$

where we have, in our special case with $s = 0, \lambda_{EE} = 1, \xi = 1, \chi = 0$ ($\mathcal{C} = 0.5$ in Appendix of Bloesch, Lee and Weber (2026)), $\epsilon_R = \frac{\gamma}{2}$ and $\epsilon_S = -\frac{\gamma}{2}$, leading to

$$\Lambda_1 = \epsilon_R - S(\epsilon_R - \epsilon_S) = \gamma\left(\frac{1}{2} - S\right)$$
$$\Delta_1 = -\epsilon_S + S(\epsilon_R - \epsilon_S) = \gamma\left(\frac{1}{2} + S\right)$$

with $\epsilon_S g_{\epsilon_S,V} = 0$. We also have that:

$$g_{S,V} = \frac{f}{f + \lambda_{EU}} \times \frac{1}{1+\theta^2}$$
$$g_{R,V} = -\frac{\theta^2}{1+\theta^2}$$

Note that with $\chi = 1 > S$ as we assume in Bloesch, Lee and Weber (2026), we could always sign $\Lambda_1 > 0$. Now with $\chi = 0$, $S > \frac{1}{2}$ would lead to $\Lambda_1 < 0$. To make progress, combine these results to write everything in terms of S and θ, noting that $f = \frac{\theta}{(1+\theta^2)^{\frac{1}{2}}}$:

$$\phi_v = \frac{\kappa}{\psi_w}\left(\underbrace{\gamma\left(\frac{1}{2} - S\right)}_{\Lambda_1} + \underbrace{\gamma\left(\frac{1}{2} + S\right)}_{\Delta_1}\left(\underbrace{\frac{f}{f+\lambda_{EU}} \times \frac{1}{1+\theta^2} + \frac{\theta^2}{1+\theta^2}}_{g_{S,V} - g_{R,V}}\right) - \underbrace{0}_{\epsilon_S g_{\epsilon_S,V}}\right)$$

which simplifies to

$$\phi_V = \frac{\kappa}{\psi_w} \cdot \gamma\left(\frac{1}{2} - S + \left(\frac{1}{2} + S\right)\frac{\frac{f}{f+\lambda_{EU}} + \theta^2}{1+\theta^2}\right). \tag{8}$$

Looking at equation (8), we can observe if the steady state labor market is sufficiently tight, i.e. θ is high so that f is high, then

$$\frac{\frac{f}{f+\lambda_{EU}} + \theta^2}{1+\theta^2} \simeq 1$$

and

$$\phi_v \simeq \frac{\kappa}{\psi_w}\gamma > 0.$$

Sufficiency Now, we note that in any steady state where $U < 50\%$, ϕ_v in equation (8) is positive.

To see this, begin by noting that

$$N = \frac{f}{f+\lambda_{EU}} \tag{9}$$

at the steady state. This follows from the law of motion for employment (6):

$$N = (1-S)N + R(V)V$$

and plugging in for $VR(V) = \frac{f(V)}{2} = S - \frac{\lambda_{EU}}{2}$, which is from equations (4) and (5). Obtaining $N = S(1-N) + N - \frac{\lambda_{EU}}{2}$, we solve for $N = 1 - \frac{\lambda_{EU}}{2S}$ and plug in for $S = \frac{1}{2}(f + \lambda_{EU})$, leading to equation (9).

Then it follows that if $N = \frac{f}{f+\lambda_{EU}} > 0.5$, i.e., $U < 0.5$, we have

$$\frac{\frac{f}{f+\lambda_{EU}} + \theta^2}{1+\theta^2} > 0.5,$$

which leads to

$$\phi_V > \frac{\kappa}{\psi_w}\gamma\left(\frac{1}{2} - S + \left(\frac{1}{2} + S\right)\frac{1}{2}\right) = \frac{\kappa}{\psi_w}\gamma\frac{1}{2}\left(1 - \underbrace{S}_{<1} + \frac{1}{2}\right) > 0$$

Finally, we can always find such a steady state by choosing the vacancy cost parameter c low so that θ and f are high in steady state, so that the steady state employment N (unemployment U) is high enough (low enough) to ensure $\phi_V > 0$.